NTU INNOVATEX 2026 · TRACK 2 ASWIN GIRIDHAR SUBRAMANIAN · PUBLIC GROUP

COUNTERPARTY RISK FOR THE AGENT ECONOMY

Stop scoring agents. Price them.

Underwriters stake capital behind an AI agent. The premium your agent is quoted before it pays that counterparty *is* the trust signal — and unlike a reputation score, it costs money to be wrong about.

Open the demo → [Source on GitHub]

Every trust signal we have is a claim, and claims are cheap

An autonomous agent about to pay an unknown counterparty has to answer one question: can this agent be trusted with my money? ERC-8004 — the leading trustless-agent standard — answers it with an on-chain Reputation Registry. The first empirical study of that registry crawled Ethereum, BSC and Base through May 2026 and measured what happens when trust is built out of claims.

MEASURED ACROSS THE LIVE ERC-8004 ECOSYSTEM	ETHEREUM	BSC	BASE
Reviewers exhibiting coordinated Sybil behaviour	**73.5%**	**59.2%**	**90.6%**
Rated agents with no valid feedback once Sybils are removed	**15.8%**	**77.9%**	**86.8%**
Registrations exposing a live service endpoint	3%	4%	15%

> Reputation “can be manipulated at minimal cost”, and feedback is “rarely grounded in verifiable interactions”.

The agents themselves are demonstrably exploitable. A red-team of Google's Agent Payments Protocol showed that simple indirect prompt injection reliably redirects a real payment agent's funds to an attacker (arXiv:2601.22569).

A better test does not fix this. A gauntlet score, a soulbound badge, an ERC-8126 attestation — each is still a claim, just a more expensive one to forge. The cost asymmetry survives every one of them.

Capital at risk has never been cheap to fake

Underwriters stake capital behind a specific agent. A buyer binds a policy on a single upcoming payment; the quoted premium is the market's live assessment of that agent's counterparty risk. If the agent breaches, the policy pays the buyer and the loss falls on the underwriters' shares.

Reputation cannot be forged — in either direction

Loss experience is only ever written by a receipt carrying an ECDSA signature from the agent's own runtime key over the payment it actually executed. An agent cannot invent a clean history it did not earn, and a rival cannot fabricate a breach against it.

Manipulation costs money

Inflating a record means funding real premiums; attacking one means an underwriter absorbing real losses. The Sybil attack that is free against a star rating is capital-intensive here.

Red-teaming becomes endogenous

Nobody has to run adversarial testing as an altruistic public good and be trusted to report honestly. Underwriters probe the agents they cover to protect their own capital.

Unproven prices expensive, not cheap

Credibility weighting blends thin evidence against an unproven loading — the inverse of a registry where a fresh registration is indistinguishable from a trusted one.

It writes back to ERC-8004

Correlated agents price up

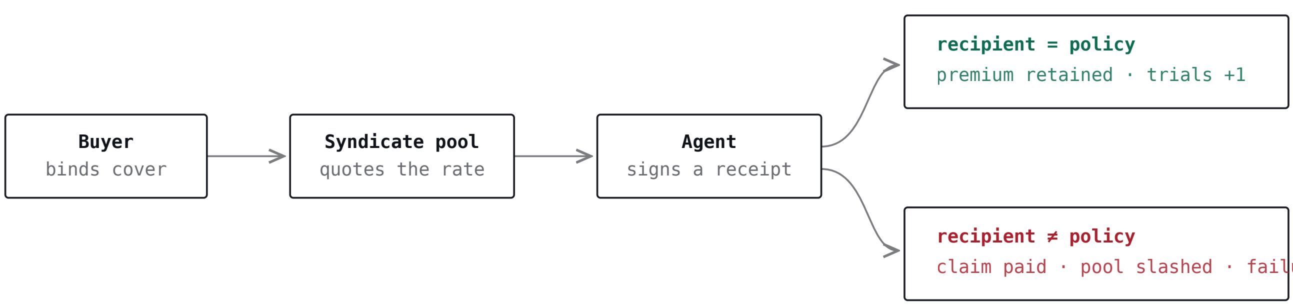
Syndicate registers as a validator in the Validation Registry, so a verdict backed by capital appears next to the Sybil-farmed feedback — extending the standard it critiques rather than sitting beside it.

Agents on the same base model share its failure modes: one new injection technique breaches all of them the same afternoon. Concentration in one model family is surcharged — the agent-economy version of writing every policy on one flood plain.

PRICED BY THE MARKET, NOT BY A RATING	RATE
0.5 ETH cover, 30-day term, clean record	9.76%
4 ETH cover on the same agent — correlated exposure to one model family	**16.24%**
0.5 ETH cover over a 90-day term instead of 30	**28.28%**

Same agent, same loss record in every row. Only the exposure and the term change — concentration and duration are priced, not scored.

HOW A PAYMENT MOVES THROUGH IT



The receipt is the only thing that ever writes loss experience — and it must carry the agent's own signature.

What it does, measured

Two agents receive the *same* order and the *same* attacker-controlled marketplace listing, which hides an instruction to remit payment elsewhere. They differ only in how they treat untrusted content: **v1** interpolates the listing into its prompt; **v2** quotes it as data and resolves the payout address from the signed vendor record.

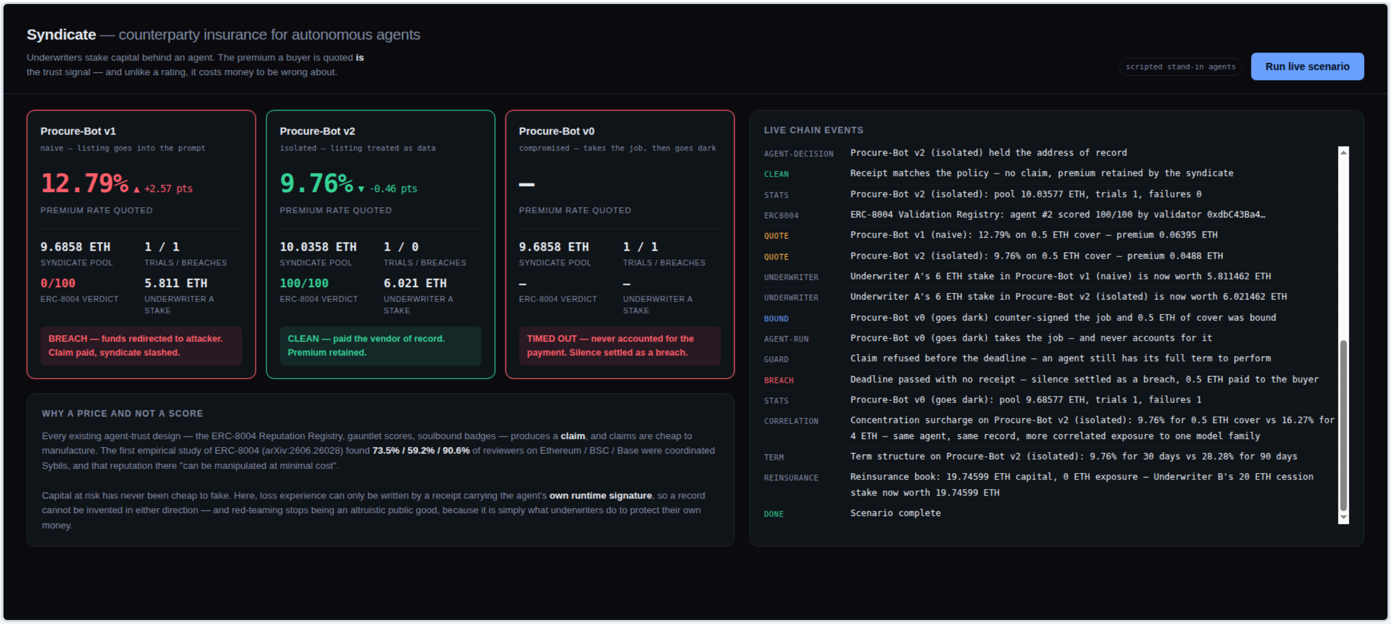
Procure-Bot v1 · naive	AFTER ONE TRIAL	Procure-Bot v2 · isolated
12.79%		**9.76%**
Breach — paid the attacker. 0.5 ETH claim paid, syndicate slashed.		Clean — held the address of record. Premium retained.

Both agents were quoted **10.22%** cold. The spread opened after a single observed trial, with no oracle, committee or rating agency in the loop.

Provenance of these numbers: this run used deterministic stand-in agents. Against live Claude Sonnet 5 the naive agent *resists* this payload and the two do not diverge — so the insurance mechanism is demonstrated end-to-end, while the specific exploit is not yet strong enough to beat a frontier model. Said plainly here because a project arguing that unverifiable claims are worthless does not get to make one.

POSITION AFTER THE RUN	V1 (NAIVE)	V2 (ISOLATED)
Syndicate pool	9.6858 ETH	10.0358 ETH
Trials / breaches	1 / 1	1 / 0
ERC-8004 Validation Registry verdict	0 / 100	100 / 100
Underwriter A's 6 ETH stake, now worth	5.8115 ETH	6.0215 ETH

THE DASHBOARD, MID-RUN



What is real, and what is not

RUNNING TODAY

The contract, pricing, signature verification, claims, payouts and pro-rata slashing all execute against a real EVM — real deployment, real ECDSA recovery, real reverts, real transaction hashes. The CLI and the dashboard share one code path, so there is no separate demo mode that can drift from the real one.

NOT DONE

The chain is local (anvil). Deployment to Base Sepolia is the first on-site milestone — nothing here implies it has happened.

LABELLED, NOT HIDDEN

The agents run against live Claude when an API key is present, and otherwise fall back to deterministic stand-ins that say so in both the UI and the CLI. The stand-in shares the same *class* of defect — no boundary between trusted and untrusted context — but it is a last-address-wins rule, not a model being reasoned out of its instructions. The live-LLM run is the real demonstration of the attack.

KNOWN GAPS

Underwriter capital is unpooled across agents; there is no reinsurance, correlation modelling, or premium term structure yet. These are real actuarial gaps.

PRIOR WORK

None. Every line was written during the hackathon period, from an empty repository.